Advanced Log Analysis

Introduction

This practical exercise focused on performing advanced log analysis using security monitoring tools. The main objective was to ingest logs into Elastic Security, analyze authentication events, detect anomalies, and enrich log data with geographic information. The tools used in this lab included Elastic Security for log analysis and monitoring, Security Onion for log collection context, and Google Sheets for documenting correlated events. The analysis focused particularly on identifying failed login attempts (Event ID 4625) and correlating them with outbound network traffic.

Tools Used

Tool	Purpose
Elastic Security	Log ingestion, analysis, and enrichment
Security Onion	Security monitoring environment
Filebeat	Log forwarding to Elasticsearch
Google Sheets	Documentation of correlated logs

Log Ingestion and Correlation

The first step involved ingesting sample log data into Elasticsearch using Filebeat. The logs were generated manually to simulate failed authentication attempts and outbound network connections. Once the logs were ingested, they were analyzed using the **Discover** feature in Elastic Security.

The objective of this stage was to identify patterns between authentication failures and outbound network traffic. Event ID **4625**, which represents a failed login attempt, was used as the primary indicator of suspicious activity.

During the analysis, four log entries were created and ingested into Elasticsearch. These logs simulated repeated authentication failures originating from the same source IP address and communicating with external DNS servers.

Correlated Log Events

The following table summarizes the correlated log events observed during the analysis.

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-03-26 12:00:00	4625	192.168.1.100	1.1.1.1	Outbound DNS connection detected
2026-03-26 12:03:00	4625	192.168.1.100	8.8.8.8	Failed login attempt observed
2026-03-26 12:05:00	4625	192.168.1.100	8.8.8.8	Repeated authentication failure
2026-03-26 12:06:00	4625	192.168.1.100	8.8.8.8	Suspicious login activity

These repeated failed login attempts indicate a possible brute-force attempt originating from the internal host **192.168.1.100**. The correlation with outbound DNS traffic suggests that the host may be communicating with external infrastructure during the authentication attempt.

Threat Analysis

The repeated failed login attempts from source IP **192.168.1.100** combined with outbound DNS communication to **8.8.8.8** and **1.1.1.1** may indicate suspicious behavior. This pattern could represent:

- Brute force authentication attempts
- Credential guessing attack
- Malware attempting external communication
- DNS queries used for command-and-control communication

Such correlations help SOC analysts identify compromised hosts or potential insider threats.

Anomaly Detection Rule

To detect abnormal network behavior, a rule was designed to identify high-volume data transfers. The rule condition was defined to trigger when outbound traffic exceeds **1 MB within one minute**.

However, during the implementation stage, the Elastic Security interface displayed a **"Detection engine permissions required"** message. This indicates that the current user account did not have sufficient privileges to create or manage detection rules in the Elastic SIEM environment.

Despite the permission restriction, the detection rule configuration was designed and documented.

Planned Rule Configuration

Parameter	Value
Rule Name	High Volume Data Transfer Detection
Index Pattern	filebeat-*
Condition	bytes_out > 1MB
Time Window	1 minute
Alert Action	Generate security alert

Steps to Create the Detection Rule

1. Navigate to **Elastic Security → Rules**.
2. Click **Create Rule**.
3. Select **Custom Query Rule**.
4. Set the index pattern to **filebeat-***.
5. Define the detection condition:

```
bytes_out > 1000000
```

6. Configure the rule to run every **1 minute**.
7. Set the alert threshold.
8. Save and activate the rule.

If permissions were available, this rule would generate alerts whenever unusually large data transfers are detected.

Log Enrichment Using GeoIP

The final stage of the exercise involved enriching log data with geographic information. This was achieved by creating an **Elasticsearch ingest pipeline** that used the **GeoIP processor** to analyze destination IP addresses.

The pipeline configuration was as follows:

```
{
 "geoip": {
   "field": "dest_ip",
   "target_field": "geoip"
 }
}
```

This configuration allows Elasticsearch to automatically retrieve geographic information associated with an IP address using the GeoLite database.

The pipeline was tested using the Elasticsearch **simulate API** in Dev Tools.

Example test command:

```
POST _ingest/pipeline/geoip_enrichment_pipeline/_simulate
{
  "docs": [
    {
      "_source": {
        "dest_ip": "8.8.8.8"
      }
    }
  ]
}
```

The result successfully returned geographic information including continent name, country name, and geographic coordinates.

Example output fields included:

- continent_name: North America
- country_name: United States
- latitude: 37.751
- longitude: -97.822

This enrichment allows analysts to quickly identify the geographic origin of network traffic.

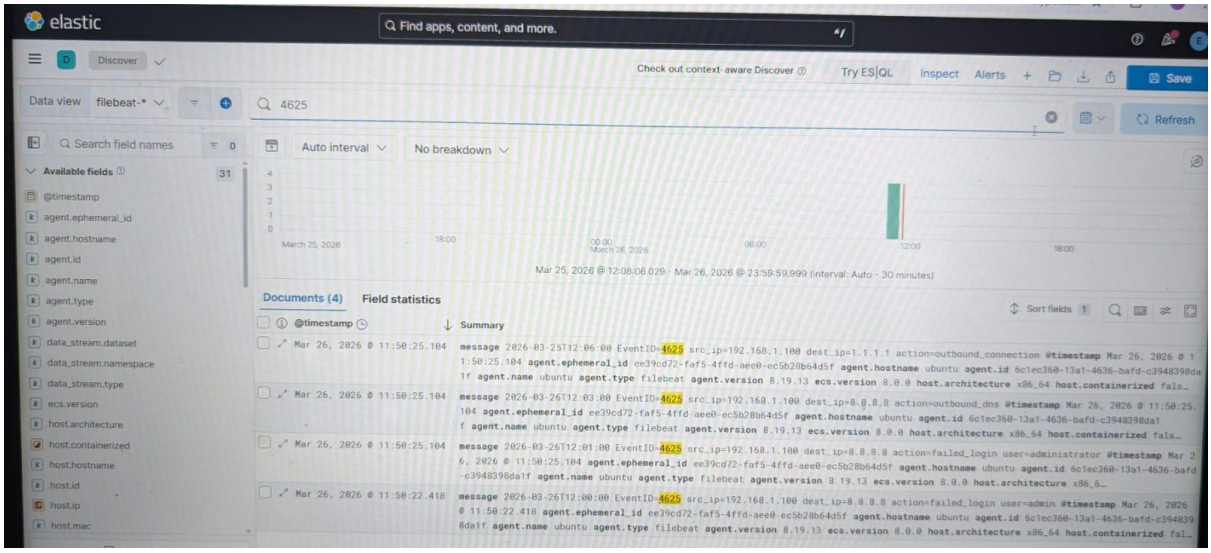
GeoIP Enrichment Summary

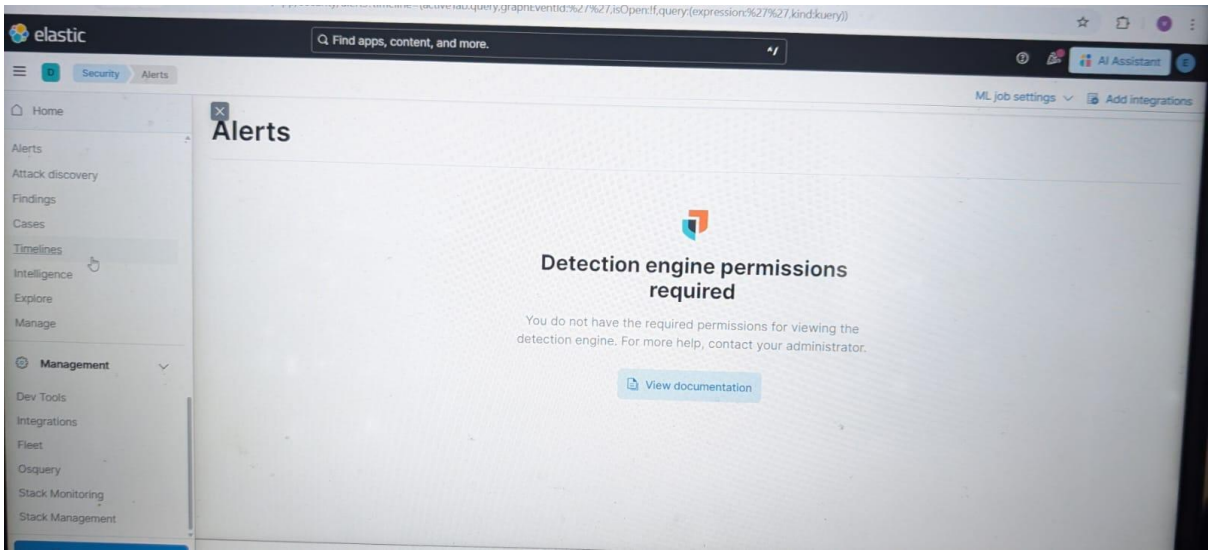
GeoIP enrichment was implemented in Elasticsearch to enhance log data with geographic information. The GeoIP processor analyzed destination IP addresses and retrieved location details such as continent, country, and coordinates. The pipeline was tested using the simulate API with IP address 8.8.8.8, confirming successful enrichment and improved contextual visibility.

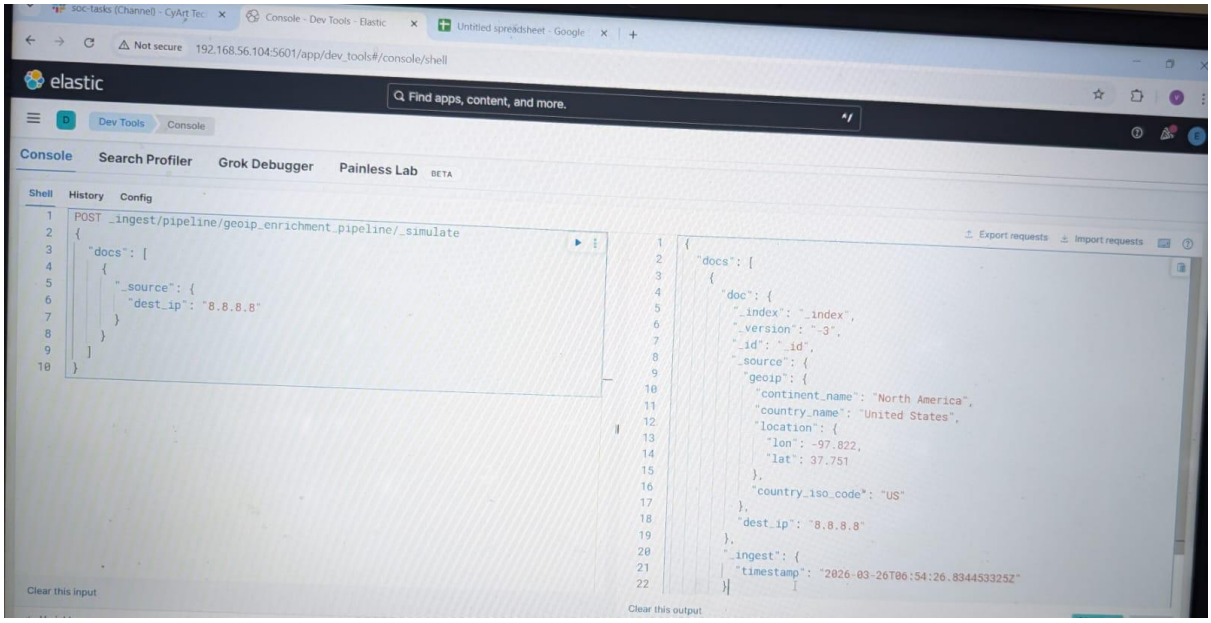
Conclusion

This practical exercise demonstrated the process of performing advanced log analysis using Elastic Security. Logs were successfully ingested and correlated to identify repeated failed authentication attempts. Although detection rules could not be deployed due to permission limitations, the rule configuration was documented. Finally, GeoIP enrichment was implemented to enhance log visibility by providing geographic context for IP addresses.

The exercise highlights the importance of log correlation, anomaly detection, and data enrichment in modern Security Operations Center (SOC) workflows.






soc-tasks (Channel) - CyArt Tec
Console - Dev Tools - Elastic
Untitled spreadsheet - Google
Not secure
192.168.56.104:5601/app/dev_tools#/console/shell
elastic
Find apps, content, and more.
Dev Tools
Console
Console
Search Profiler
Grok Debugger
Painless Lab
BETA
Shell
History
Config
POST _ingest/pipeline/geoip_enrichment_pipeline/_simulate
{
"docs": [
{
"_source": {
"dest_ip": "8.8.8.8"
}
}
]
}
Export requests
Import requests
{
"docs": [
{
"doc": {
"_index": "_index",
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"_source": {
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"lat": 37.751
},
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},
"dest_ip": "8.8.8.8"
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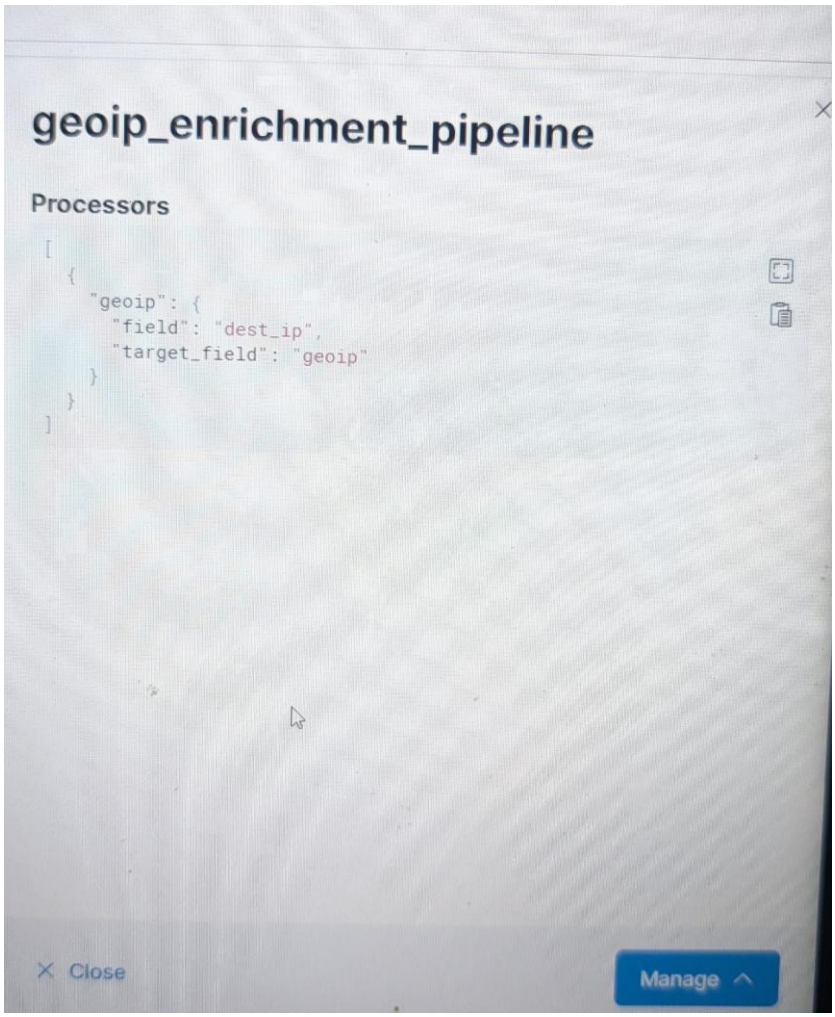
}
Clear this input
Clear this output


geoip_enrichment_pipeline
Processors
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{
"geoip": {
"field": "dest_ip",
"target_field": "geoip"
}
}
]
Close
Manage